# High-nickel tree nuts (walnuts, pine nuts, Brazil nuts, chestnuts): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified standard for High-nickel tree nuts (walnuts, pine nuts, Brazil nuts, chestnuts) (Nuts, Seeds, and Their Butters), stated as sold. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; regulatory-alignment / read-across / not-material / control)

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 200 | government maximum level |
| Cadmium (Cd) | Tier 1 | 200 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 800 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |

| | | | |
|---|---|---|---|
| Mercury, total (tHg) | Tier 1 | 400 | human-grade read-across (Israel) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 10,000 | government maximum level |
| Tin (Sn) | Tier 2 | 250,000 | human-grade read-across (Singapore) |
| Aluminium (Al) | Tier 2 | 100,000 | human-grade read-across (China) |
| Total chromium (Cr) | Tier 2 | 1,000 | human-grade read-across (China) |
| Hexavalent chromium (Cr-VI) | Tier 2 | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 200 | government maximum level |
| Cadmium (Cd) | 200 | government maximum level |
| Total arsenic (tAs) | 800 | government maximum level |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 400 | Israel — human-food analog |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 10,000 | government maximum level |
| Tin (Sn) | 250,000 | Singapore — human-food analog |
| Aluminium (Al) | 100,000 | China — human-food analog |

| | | |
|---|---|---|
| Total chromium (Cr) | 1,000 | China — human-food analog |
| Hexavalent chromium (Cr-VI) | ≤ Cr | Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell. |

## 4 | Cite this standard

Pendergrass, K. "High-nickel tree nuts (walnuts, pine nuts, Brazil nuts, chestnuts)." In Heavy Metal Tested & Certified (HMTc) Nuts, Seeds, and Their Butters standard, version 1.0 (2026-08-14). Institute of Contaminant Standards (ICS), 2026.

## References


[1] China — human-food maximum levels
Binds Al, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Israel — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL


### STATUS AND LICENCE

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### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/tree--nuts-high-nickel. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

### PUBLICATION HISTORY

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